

## SIMUSIR, Russian Federation

WMO: 321950

Lat: 46.850N

Lon: 151.867E

Elev: 25

StdP: 101.03

Time Zone: 10.00 (E10)

Period: 86-97

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.7       | -8.6 | -14.5                         | 1.1 | -8.6 | -12.8 | 1.3 | -7.9 | 27.6                  | -3.4 | 24.0 | -4.7 | 10.1                  | 290  | 0.964 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.5                    | 19.1            | 16.7 | 16.8 | 14.7 | 15.0 | 13.5 | 17.3                | 18.7 | 15.3 | 16.4 | 13.7 | 14.6 | 7.3                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.6                            | 11.9 | 18.5 | 14.8 | 10.6 | 16.0 | 13.2 | 9.5 | 14.3 | 49.1          | 18.8 | 43.2 | 16.5 | 38.7 | 14.7 | 23.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 21.6          | 18.6 | 16.5 | -11.4                      | 24.0 | 1.6                | 2.1 | -12.6                                              | 25.5 | -13.5      | 26.7 | -14.4      | 27.9 | -15.6      | 29.4 |    |    |
| (5)               |      |      | -11.7                      | 20.9 | 1.8                | 1.8 | -13.1                                              | 22.3 | -14.1      | 23.3 | -15.1      | 24.4 | -16.4      | 25.7 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.2   | -4.0  | -3.9  | -2.3  | 1.0   | 3.8   | 5.6   | 8.2   | 10.9  | 10.3  | 7.4   | 2.3   | -1.2 | (6)  |
| (7)  |                                                                                  | DBStd       | 5.73  | 2.67  | 2.73  | 2.22  | 2.44  | 2.51  | 2.31  | 2.60  | 2.92  | 2.42  | 2.38  | 2.81  | 2.59 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2579  | 433   | 389   | 383   | 269   | 192   | 136   | 68    | 21    | 23    | 88    | 231   | 347  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5518  | 691   | 622   | 641   | 519   | 449   | 383   | 313   | 231   | 241   | 339   | 481   | 606  | (9)  |
| (10) |                                                                                  | CDD10.0     | 104   | 0     | 0     | 0     | 0     | 1     | 2     | 14    | 49    | 33    | 6     | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 2     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.5   | 9.9   | 9.9   | 9.4   | 8.5   | 6.6   | 5.3   | 4.8   | 5.1   | 5.8   | 7.4   | 8.2   | 9.2  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1365  | 93    | 82    | 95    | 100   | 97    | 87    | 112   | 131   | 157   | 148   | 143   | 121  | (15) |
| (16) |                                                                                  | PrecMax     | 1706  | 144   | 204   | 178   | 181   | 148   | 156   | 225   | 306   | 254   | 234   | 296   | 191  | (16) |
| (17) |                                                                                  | PrecMin     | 1051  | 51    | 38    | 46    | 45    | 41    | 34    | 36    | 39    | 72    | 69    | 65    | 75   | (17) |
| (18) |                                                                                  | PrecStd     | 168   | 28    | 32    | 30    | 32    | 26    | 29    | 42    | 59    | 45    | 40    | 49    | 28   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 2.3   | 2.4   | 4.6   | 11.4  | 16.1  | 15.7  | 20.1  | 24.0  | 22.4  | 16.3  | 12.4  | 7.8  | (19) |
| (20) |                                                                                  |             | MCWB  | 1.3   | 1.6   | 2.8   | 7.5   | 10.0  | 12.9  | 17.6  | 20.6  | 20.6  | 14.1  | 10.4  | 6.8  | (20) |
| (21) |                                                                                  | 2%          | DB    | 1.5   | 1.3   | 2.5   | 7.8   | 12.3  | 12.5  | 16.9  | 20.1  | 17.8  | 13.9  | 9.1   | 5.4  | (21) |
| (22) |                                                                                  |             | MCWB  | 0.7   | 0.6   | 1.3   | 5.6   | 8.7   | 10.3  | 15.1  | 18.1  | 15.4  | 12.0  | 7.9   | 4.7  | (22) |
| (23) |                                                                                  | 5%          | DB    | 1.0   | 0.7   | 1.4   | 5.4   | 10.0  | 10.4  | 14.2  | 17.4  | 15.6  | 12.3  | 7.8   | 3.9  | (23) |
| (24) |                                                                                  |             | MCWB  | 0.3   | 0.0   | 0.6   | 3.8   | 6.9   | 9.0   | 12.9  | 15.9  | 13.4  | 10.4  | 6.9   | 3.2  | (24) |
| (25) |                                                                                  | 10%         | DB    | 0.3   | 0.1   | 0.6   | 4.1   | 7.8   | 8.8   | 11.9  | 15.3  | 13.9  | 10.9  | 6.6   | 2.2  | (25) |
| (26) |                                                                                  |             | MCWB  | -0.4  | -0.6  | -0.1  | 3.0   | 5.9   | 7.8   | 11.1  | 14.4  | 12.4  | 9.4   | 5.8   | 1.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 1.7   | 2.0   | 3.4   | 8.2   | 11.8  | 13.6  | 17.9  | 20.7  | 20.6  | 14.2  | 10.9  | 6.9  | (27) |
| (28) |                                                                                  |             | MCDWB | 2.0   | 2.2   | 4.3   | 10.9  | 13.8  | 15.1  | 19.3  | 22.3  | 22.5  | 15.8  | 11.9  | 7.4  | (28) |
| (29) |                                                                                  | 2%          | WB    | 0.9   | 0.8   | 1.7   | 5.6   | 8.8   | 10.8  | 15.5  | 18.2  | 15.9  | 12.4  | 8.4   | 5.0  | (29) |
| (30) |                                                                                  |             | MCDWB | 1.3   | 1.1   | 2.3   | 7.4   | 11.5  | 12.2  | 16.8  | 19.9  | 17.2  | 13.6  | 9.0   | 5.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | 0.4   | 0.1   | 0.8   | 4.3   | 7.1   | 9.2   | 13.1  | 16.1  | 13.9  | 11.0  | 7.1   | 3.1  | (31) |
| (32) |                                                                                  |             | MCDWB | 0.9   | 0.5   | 1.3   | 5.3   | 9.2   | 10.2  | 14.1  | 16.9  | 14.9  | 11.9  | 7.6   | 3.7  | (32) |
| (33) |                                                                                  | 10%         | WB    | -0.4  | -0.5  | 0.0   | 3.1   | 5.8   | 7.8   | 11.2  | 14.4  | 12.5  | 9.7   | 5.7   | 1.5  | (33) |
| (34) |                                                                                  |             | MCDWB | 0.3   | 0.0   | 0.5   | 3.9   | 7.3   | 8.5   | 11.7  | 15.2  | 13.6  | 10.5  | 6.3   | 2.1  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 3.0   | 2.7   | 3.0   | 3.6   | 5.0   | 4.5   | 4.6   | 5.5   | 5.2   | 4.6   | 3.8   | 3.2  | (35) |
| (36) |                                                                                  |             | MCDBR | 3.3   | 2.9   | 4.1   | 7.3   | 10.4  | 8.4   | 9.1   | 9.2   | 8.3   | 6.9   | 6.0   | 5.7  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.4   | 2.9   | 4.1   | 5.4   | 6.7   | 6.4   | 7.3   | 7.6   | 6.3   | 5.7   | 6.1   | 5.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 3.6   | 3.0   | 4.1   | 7.1   | 8.8   | 8.0   | 8.7   | 9.0   | 7.8   | 6.6   | 6.0   | 5.5  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.6   | 3.1   | 4.1   | 5.5   | 6.6   | 6.3   | 7.3   | 7.5   | 6.7   | 5.8   | 6.2   | 5.6  | (39) |
|      |                                                                                  |             |       |       |       |       |       |       |       |       |       |       |       |       |      |      |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.360 | 0.382 | 0.443 | 0.492 | 0.502 | 0.484 | 0.466 | 0.440 | 0.394 | 0.403 | 0.393 | 0.370 | (40) |      |
| (41) |                                                                                  | taud        | 2.117 | 2.076 | 1.962 | 1.894 | 1.942 | 2.074 | 2.201 | 2.285 | 2.372 | 2.168 | 2.098 | 2.066 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 680   | 753   | 759   | 763   | 773   | 791   | 800   | 807   | 815   | 723   | 630   | 606   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 93    | 119   | 155   | 183   | 180   | 159   | 139   | 122   | 103   | 108   | 95    | 88    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.08  | 1.77  | 2.90  | 3.75  | 4.12  | 4.25  | 4.08  | 3.60  | 3.31  | 2.38  | 1.28  | 0.90  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.07  | 0.14  | 0.25  | 0.43  | 0.49  | 0.36  | 0.39  | 0.27  | 0.29  | 0.17  | 0.12  | 0.05  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page